

Sleep & Energy Analysis

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Intro

In this experiment, I tracked my sleep, reaction, and daily energy over a 14-day period. The first week (10/11-10/17) represents my controlled week, where I tried to obtain 8 hours of sleep per night, and Week two (10/18/10/24) represents my regular sleep.

Contextual Notes

Gathering the controlled data was especially difficult, getting a full 8 hours of sleep was especially hard, and I failed to do so on some days. My fix to this was collecting data on easier days, like the weekend, then using that data in place for the days that I failed to gather accurate, controlled data. The majority of the issues I ran into were scheduling, looking at the drastic differences between my regular days and controlled days, it makes sense that this would be an issue for me. Sleeping and waking up at the same time was extremely unsustainable, and I had to resort to such a method to have data for these days.

There was also one day in which the Apple Watch made an error in tracking and reported that I slept for 17 hours; I had to manually go into the health data app to fix the error.

##Data Gathering

```
#Read in the data
sleep_df <- read_excel("Sleep_Energy_data.xlsx")

names(sleep_df)

## [1] "Date"           "WeekType"       "Bedtime"
## [4] "Wake Time"      "Time In Bed (hrs)" "Total Sleep (hrs)"
## [7] "Core (hrs)"     "REM (hrs)"      "Deep (hrs)"
## [10] "Awake (hrs)"    "Sleep Efficiency" "EnergyFocus"
## [13] "RT_Morning_ms"  "RT_Evening_ms"   "RT_Best_ms"
## [16] "RT_Worst_ms"    "RT_Mean_ms"

# Clean column types
sleep_df <- sleep_df %>%
  mutate(
    Date = as.Date(Date),
    WeekType = factor(WeekType, levels = c("Control", "Regular"))
  )

# Print cleaned data
sleep_df
```

```
## # A tibble: 14 × 17
##   Date       WeekType Bedtime       `Wake Time`       `Time In Bed
(hrs)`
##   <date>      <fct>    <chr>          <chr>
<dbl>
## 1 2025-10-11 Control  2025-10-10 23:12 2025-10-11 07:00
7.83
## 2 2025-10-12 Control  2025-10-11 23:24 2025-10-12 07:00
7.63
## 3 2025-10-13 Control  2025-10-12 23:00 2025-10-13 07:00
8.03
## 4 2025-10-14 Control  2025-10-13 23:30 2025-10-14 07:00
7.53
## 5 2025-10-15 Control  2025-10-14 23:06 2025-10-15 07:00
7.93
## 6 2025-10-16 Control  2025-10-15 23:18 2025-10-16 07:00
7.73
## 7 2025-10-17 Control  2025-10-16 22:54 2025-10-17 07:00
8.13
## 8 2025-10-18 Regular 02:13          09:34
7.35
## 9 2025-10-19 Regular 00:30          08:34
8.06
## 10 2025-10-20 Regular 03:34          07:58
4.39
## 11 2025-10-21 Regular 05:40          13:38
7.97
## 12 2025-10-22 Regular 01:21          09:45
8.4
## 13 2025-10-23 Regular 03:25          08:52
5.45
## 14 2025-10-24 Regular 03:20          08:10
4.83
## # i 12 more variables: `Total Sleep (hrs)` <dbl>, `Core (hrs)` <dbl>,
## #   `REM (hrs)` <dbl>, `Deep (hrs)` <dbl>, `Awake (hrs)` <dbl>,
## #   `Sleep Efficiency` <dbl>, EnergyFocus <dbl>, RT_Morning_ms <dbl>,
## #   RT_Evening_ms <dbl>, RT_Best_ms <dbl>, RT_Worst_ms <dbl>, RT_Mean_ms
<dbl>
```

##Visualization + Interpretation

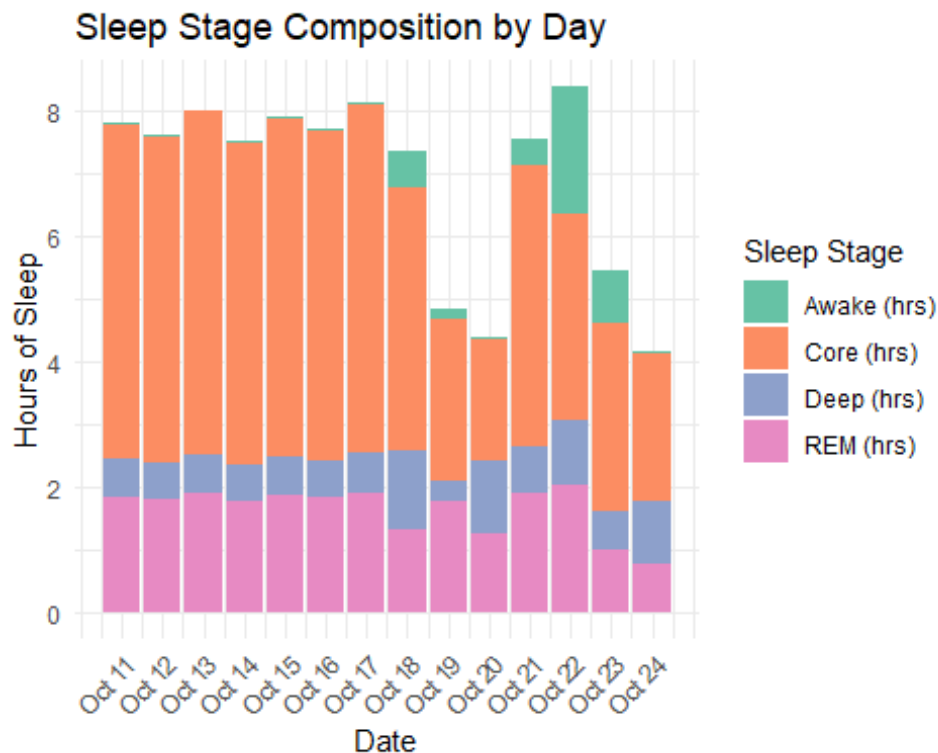
#Sleep Data

```
ggplot(sleep_long, aes(Date, Hours, fill = Stage)) +
  geom_bar(stat = "identity") +
  scale_x_date(
    breaks = unique(sleep_long$Date),
    date_labels = "%b %d"
  ) +
  labs(
```

```

title = "Sleep Stage Composition by Day",
x = "Date",
y = "Hours of Sleep",
fill = "Sleep Stage"
) +
theme_minimal() +
scale_fill_brewer(palette = "Set2") +
theme(
  axis.text.x = element_text(angle = 45, hjust = 1)  #tilt text
)

```



I chose this visualization because it presents all the sleep-stage data collected from my Apple Watch in a clear and organized way, similar to how it appears in the Health app. The graph helps show not just how long I slept each night, but the quality of that sleep based on the distribution of different sleep stages.

The graph displays:

Awake (green): Short interruptions throughout the night where the body briefly wakes up.

REM (pink): A key stage in quality sleep, associated with dreaming and memory processing. Research shows that REM sleep in the first 90 minutes is important for cognitive recovery.

Core (orange): The stage where muscle activity decreases and body temperature drops. This is often considered “light sleep,” but it makes up the majority of total sleep and is essential for overall rest.

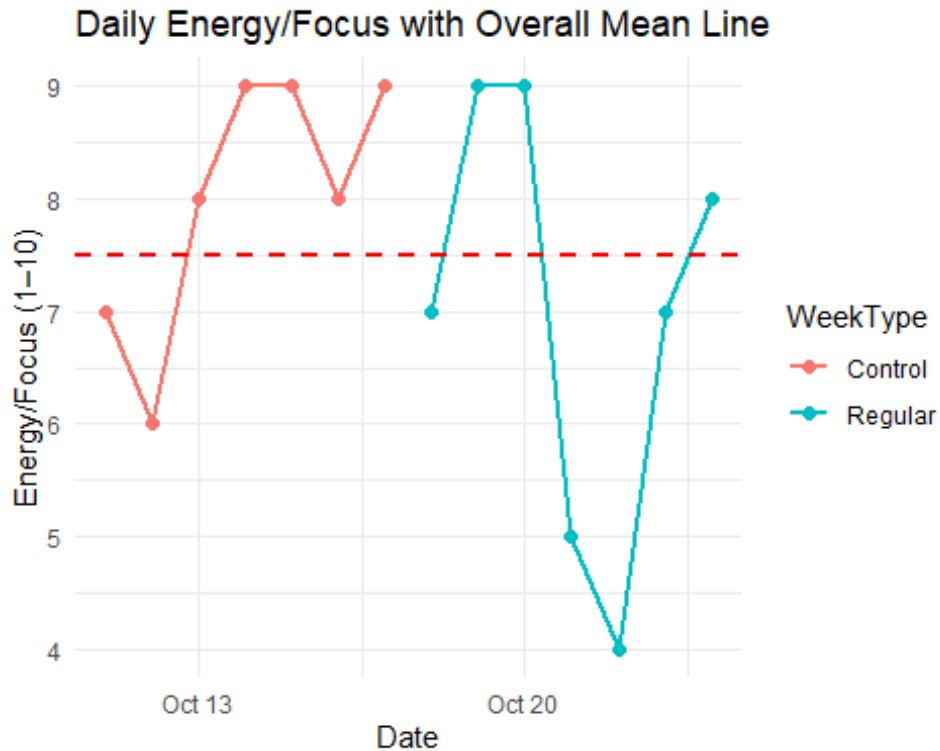
Deep (blue): The most restorative stage of sleep, where the body repairs tissue, strengthens the immune system, and releases important hormones.

One noticeable pattern in the visualization is that deep sleep remained low across both weeks, indicating that even on nights with longer total sleep, the most restorative stage was still limited.

#Energy/Focus Trend Lines Overtime

```
ggplot(sleep_df, aes(Date, EnergyFocus, color = WeekType)) +  
  geom_line(size = 1) +  
  geom_point(size = 2) +  
  geom_hline(  
    yintercept = mean(sleep_df$EnergyFocus),  
    linetype = "dashed",  
    color = "red",  
    size = 1  
  ) +  
  labs(title = "Daily Energy/Focus with Overall Mean Line",  
        x = "Date",  
        y = "Energy/Focus (1-10)") +  
  theme_minimal()
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use `linewidth` instead.  
## This warning is displayed once every 8 hours.  
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was  
## generated.
```



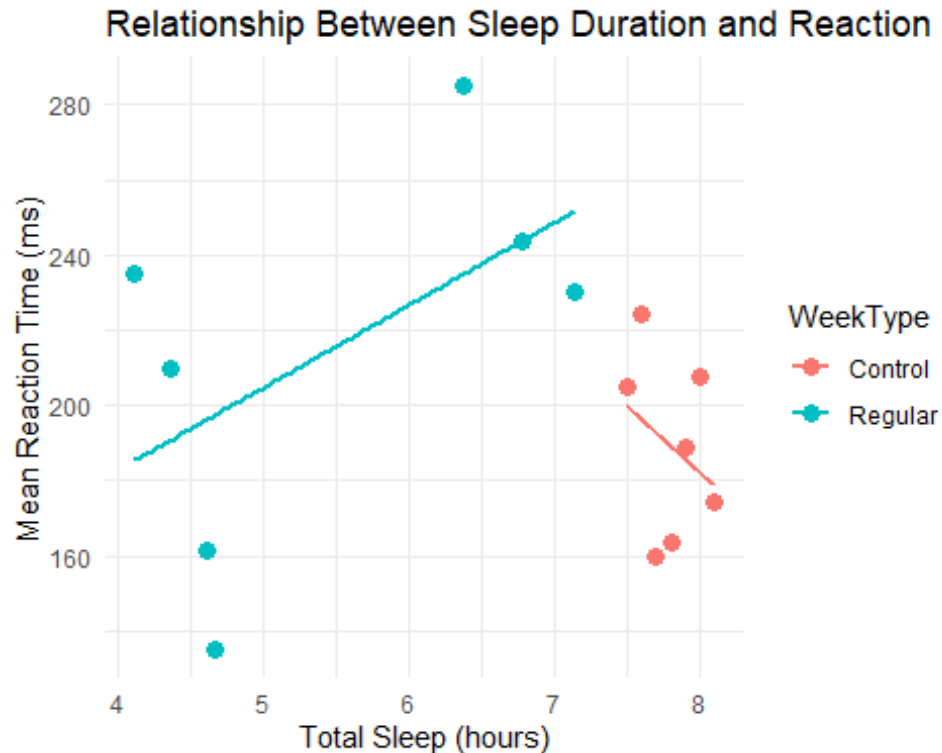
I chose this

visualization because it clearly highlights the difference in daily energy levels between the two test weeks. The addition of the overall mean line makes it easier to compare each day to the average and quickly identify any noticeable dips or peaks in energy. This helps show whether the controlled sleep week resulted in higher or more stable energy compared to my regular week

The controlled week shows consistently higher energy levels, with daily ratings clustered around 8–9. In contrast, the regular week displays a noticeable decline in energy, including several sharp drops between October 20–22. This suggests that inconsistent or shorter sleep during the regular week may have contributed to reduced daily focus and motivation. The contrast between the two lines supports the idea that maintaining a structured sleep schedule can lead to more stable and higher daytime energy.

```
ggplot(sleep_df, aes(x = `Total Sleep (hrs)`, y = RT_Mean_ms, color = WeekType)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Relationship Between Sleep Duration and Reaction Time",
    x = "Total Sleep (hours)",
    y = "Mean Reaction Time (ms)"
  ) +
  theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'
```



This visualization attempts to identify a positive or negative correlation between sleep hours and reaction time. A downward-sloping trend line would indicate that more sleep is associated with faster reaction times (lower milliseconds), while an upward-sloping line would suggest the opposite. In the plot, the control week shows a slight downward trend, meaning longer sleep tends to align with quicker reaction times. This fits my expectation that consistent rest may support better responsiveness.

During the Regular week, the data is more scattered, with points around the 5 hour sleep range showing the fastest reaction times. These outliers suggest factors beside sleep may have also influenced performance. So overall the visualization shows a weak but noticeable pattern.

##Statistical Analysis

Energy/Focus: Control vs Regular

```
t_energy <- t.test(EnergyFocus ~ WeekType, data = sleep_df)
t_energy
```

```
##
```

```
## Welch Two Sample t-test
```

```
##
```

```
## data: EnergyFocus by WeekType
```

```
## t = 1.1832, df = 9.854, p-value = 0.2645
```

```
## alternative hypothesis: true difference in means between group Control and group Regular is not equal to 0
```

```
## 95 percent confidence interval:
```

```
## -0.8869094 2.8869094
```

```

## sample estimates:
## mean in group Control mean in group Regular
##                8                7

# Reaction Time: Control vs Regular
t_rt <- t.test(RT_Mean_ms ~ WeekType, data = sleep_df)
t_rt

##
##  Welch Two Sample t-test
##
## data:  RT_Mean_ms by WeekType
## t = -1.1794, df = 8.6094, p-value = 0.2698
## alternative hypothesis: true difference in means between group Control and
## group Regular is not equal to 0
## 95 percent confidence interval:
##  -73.91225  23.48368
## sample estimates:
## mean in group Control mean in group Regular
##                189.0714                214.2857

# Helpful summary table of group means
group_summary <- sleep_df %>%
  group_by(WeekType) %>%
  summarise(
    mean_sleep = mean(`Total Sleep (hrs)`),
    mean_energy = mean(EnergyFocus),
    mean_rt = mean(RT_Mean_ms),
    n_days = n()
  )

group_summary

## # A tibble: 2 × 5
##   WeekType mean_sleep mean_energy mean_rt n_days
##   <fct>      <dbl>      <dbl>   <dbl>   <int>
## 1 Control      7.8        8     189.     7
## 2 Regular     5.44       7     214.     7

cor_test_result <- cor.test(sleep_df$`Total Sleep (hrs)`,
  sleep_df$RT_Mean_ms)
cor_test_result

##
##  Pearson's product-moment correlation
##
## data:  sleep_df$`Total Sleep (hrs)` and sleep_df$RT_Mean_ms
## t = -0.021759, df = 12, p-value = 0.983
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.5350775  0.5260514

```

```
## sample estimates:  
##          cor  
## -0.006281295
```

#Statistical Interpretation

During the controlled week, where I aimed for consistent 8-hour nights, my mean sleep duration was 7.8 hours, compared to only 5.44 hours in the regular week. The controlled week also had a faster average reaction time (189.1 ms) compared to 214.3 ms during the regular week, suggesting that more consistent sleep may have supported better cognitive performance. Energy levels while close also trended higher in the controlled week (mean of 8) than the regular week (mean of 7).

The P-value test indicated there was little to no evidence of a strong relationship between sleep duration and reaction time. While this doesn't contradict the differences observed it does show atleast in this day to day analysis that simply sleeping more on a given night did not reliably improve reaction time. This was likely due to the irregular time that varied when testing reaction time. If the experiment were repeated with reaction-time tests taken at the same time each morning, I would likely observe clearer variation and potentially a stronger relationship between sleep and cognitive performance.

##Reflection

Tracking my sleep, energy, and reaction time over two weeks helped me identify how my bedtime routine can impact various other aspects of my life. I also learned that even though the controlled week produced noticeably higher sleep quality and better reaction-time averages, day-to-day measurements were far more variable than I expected. This made me more aware of how difficult it is to isolate one factor, like sleep, when so many daily influences can affect cognitive performance.

If I were to repeat this experiment, I would make Three major changes. First, I would standardize the reaction-time tests by completing them at the same time each morning, shortly after waking up, to reduce the noise caused by testing at different times of day. Second, I would add more consistent logging of factors such as exercise timing, and emotional state so that I could better understand which of these might be confounding the relationship between sleep and alertness. Lastly I would extend the duration of the test slightly to allow for more data to be collected specifically on the controlled section. By adding more time I would be able to collect data for the 8 hour sleep interval without added stress on me to put my daily tasks aside. The limitations of this experiment primarily stem from inconsistent data collection and the small sample size. Several measurements—especially reaction times were taken irregularly, which weakens the reliability of the results. Additionally, one week of “controlled” sleep is not enough to eliminate outside influences such as stress, school workload, or social factors that may have affected both energy levels and cognitive performance. Despite these limitations, the experiment still provided meaningful insight into my personal sleep patterns and highlighted the

importance of controlling environmental variables when evaluating the effects of sleep on mental performance.